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Title of the Project

Employee Management System (EMS) in ASP.NET

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1. Introduction

The **Human Resource Management System (HRMS)** is a web-based application designed to streamline and automate various human resource-related tasks within an organization. The system is built using **ASP.NET**, which allows for a robust, secure, and scalable solution that can handle a wide range of HR functionalities. In traditional HR processes, managing employee data, attendance, leave, payroll, and performance evaluation requires significant manual effort, often leading to inefficiencies and errors.

The HRMS aims to address these issues by providing a centralized platform where all essential HR functions can be managed electronically.

2. Objectives

- Automate employee record management to reduce manual data entry and errors.
- Track employee attendance and leave requests efficiently with real-time updates.
- Manage payroll processing accurately based on attendance, leave, and performance.
- Facilitate performance tracking and appraisals to support employee development.
- Enhance communication between the HR department and employees through a centralized system.

3. Industry Visit

As part of the research and practical exploration phase of our Employee Management System project, we visited **Arafat Paints Company Ltd.**, a local manufacturing firm situated in **Kali Bari, Barisal**. The purpose of this industry visit was to gain real-world insight into how employee data is managed in a functioning business environment and to better understand the challenges organizations face with workforce administration.



3.1. Interviewing

- Conducted interviews during the visit to **Arafat Paints Company Ltd.** with key personnel across multiple departments.
- Interviewed individuals from:
 - Human Resources Department
 - Production Supervision
 - Payroll and Accounts
- Key insights from the HR Department:
 - Employee records are maintained manually (paper files, spreadsheets).
 - Attendance, leave, and performance evaluations are tracked without automation.
- From Payroll staff:
 - Salary calculation and overtime tracking are done manually.
 - Faced issues with delays and errors in payment processing.
- From Production Supervisor:
 - Difficult to access real-time employee availability and leave status.
 - Shift management is done using basic manual methods.
- Conclusion:
 - Strong need for a centralized, automated Employee Management System.
 - Real-world feedback directly influenced the features we prioritized in our project.

4. Scopes

- The Employee Management System (EMS) is designed to serve **small to medium-sized organizations** looking to digitize and automate their employee operations.
- It will allow **HR departments and management teams** to manage employee information more efficiently.
- The system will support **multiple user roles**, such as Admin, HR Manager, and Employee, each with role-based access control.
- Features include:
 - Employee registration and record management
 - Attendance tracking and leave application handling
 - Payroll calculation and salary slip generation
 - Performance evaluation and appraisal management

- Department-wise employee categorization
- The system is intended to be **modular**, so new features like asset tracking or shift planning can be added in future updates.
- It can be deployed for **local use** on a company's internal server or as a **web-based application** accessible remotely.
- The solution aims to reduce paperwork, improve data accuracy, and save time in administrative tasks.

5. Problem Statement

In many small to mid-sized companies, employee information is still managed using paper files, spreadsheets, or outdated systems. This manual process often causes problems like data duplication, difficulty in tracking attendance and leave, and errors in payroll calculations. Communication between departments becomes slow, and accessing employee records can be time-consuming. These issues not only waste time but also reduce accuracy and productivity. To solve these problems, there is a need for a centralized, user-friendly, and automated Employee Management System that can streamline all HR activities and improve efficiency across the organization.

5.1 Target Audience

The target users of this system are small and medium-sized businesses that want to modernize their employee management process. It is especially useful for HR departments, company managers, and business owners who need better tools to manage employee data, monitor attendance, handle payroll, and track performance. This system can also help organizations with multiple departments or locations by providing centralized control and easy access to real-time employee information.

6. Requirements

To successfully develop and implement the Employee Management System, certain functional and non-functional requirements must be considered.

Functional requirements include the basic operations that the system must perform. The system should allow administrators to add, update, and delete employee records. It must support employee login, attendance tracking, leave applications, payroll generation, and performance evaluations. Each user should be able to access features based on their role,

such as admin, HR, or employee. The system should also generate reports and maintain a secure database of employee information.

Non-functional requirements focus on the overall quality and performance of the system. The platform must be easy to use, with a simple and intuitive user interface. It should be responsive, working smoothly across desktops and mobile devices. Security is a major requirement—the system must protect user data with authentication and authorization mechanisms. Additionally, the system should be reliable, with minimal downtime, and be designed in a way that allows it to be scaled in the future if needed.

7. Technology

- **Frontend:** HTML, CSS, JavaScript — to create interactive and responsive user interfaces that work across devices.
- **Backend:** ASP.NET (C#) — for implementing server-side logic, handling user requests, and managing core business functionalities.
- **Database:** SQL Server — to store, organize, and secure employee data such as profiles, attendance, payroll, and leaves.
- **Tools and Frameworks:**
 - **Visual Studio** — Used as the primary Integrated Development Environment (IDE) for coding, testing, and debugging.
 - **Entity Framework** — An Object-Relational Mapping (ORM) tool used to simplify database operations with LINQ and model-based access.
 - **Git** — Version control system for tracking code changes, enabling collaboration, and managing project history efficiently.

8. Architecture

The architecture of the Employee Management System is designed to ensure modularity, maintainability, and scalability. By using a layered and structured approach, the system separates core functionalities into distinct components, making development and future updates more manageable.

8.1 System Overview

The system is divided into three main layers: the **presentation layer**, the **business logic layer**, and the **data access layer**. The presentation layer handles user interactions, the business logic layer processes data and enforces rules, and the data access layer manages

all communication with the database. This separation allows each part of the system to function independently while still working together seamlessly.

8.2 MVC Architecture

The project is built using the **Model-View-Controller (MVC)** architecture provided by the ASP.NET framework.

- **Model:** Represents the data and business logic. It includes classes that define employee details, attendance records, payroll information, and other core entities.
- **View:** Handles the user interface. It consists of HTML, CSS, and Razor syntax that displays data and captures user input.
- **Controller:** Acts as the intermediary between the Model and the View. It processes incoming requests, performs operations using the Model, and returns appropriate Views to the user.

Using MVC allows for better code organization, easier debugging, and parallel development, as each component can be worked on independently.

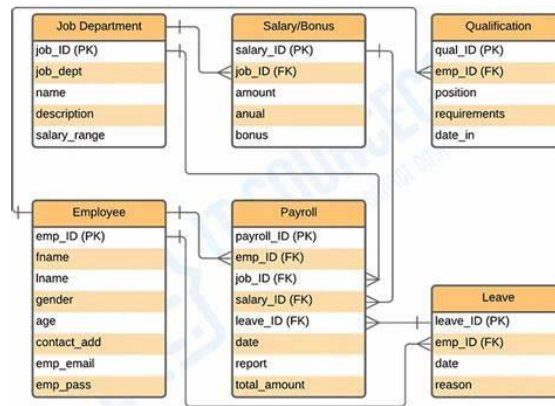
9. System Design & Visualizations

To better understand and communicate the structure and workflow of the Employee Management System, several design diagrams and visual tools were created. These visuals help in identifying system components, data flow, relationships, and timelines, making it easier for both developers and stakeholders to grasp the technical and functional aspects of the project.

9.1 ER Diagram

The **Entity-Relationship (ER) Diagram** illustrates the relationships between different entities within the system. Major entities include Employee, Department, Attendance, Leave, Salary, and Admin. The diagram helps define how data is structured and linked—such as how one department can have multiple employees or how each employee can have multiple attendance records.

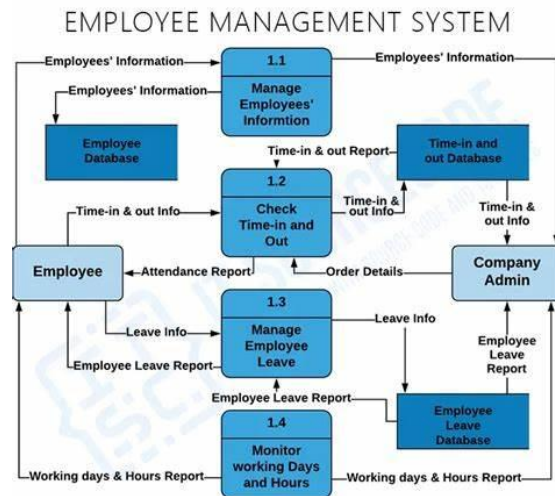
EMPLOYEE MANAGEMENT SYSTEM



ENTITY RELATIONSHIP DIAGRAM

9.2 Data Flow Diagram

The **Data Flow Diagram (DFD)** shows how data moves throughout the system. It visualizes the flow of information between users (e.g., admin or employees) and system processes such as login, payroll processing, leave approval, and report generation. This helps in understanding how the system responds to user actions and where data is stored or retrieved.

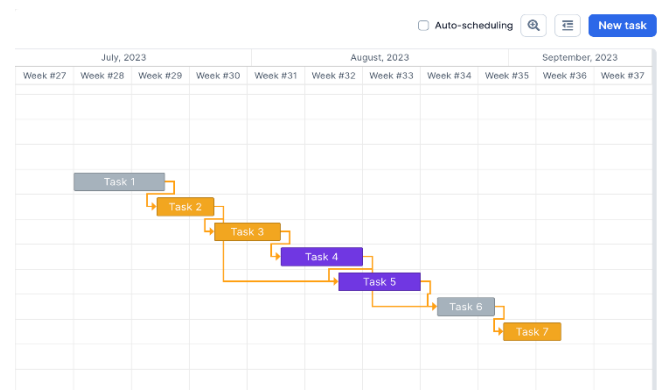
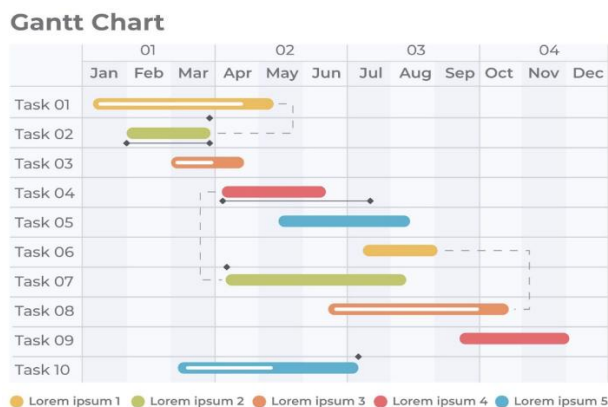


DATA FLOW DIAGRAM LEVEL 2

9.3 Gantt Chart

A **Gantt Chart** was created to plan and track the development timeline of the project. It outlines tasks such as requirement analysis, system design, backend development,

frontend development, testing, and deployment. The chart includes estimated durations and deadlines for each task, ensuring proper time management and progress monitoring throughout the development cycle.



10. Implementation

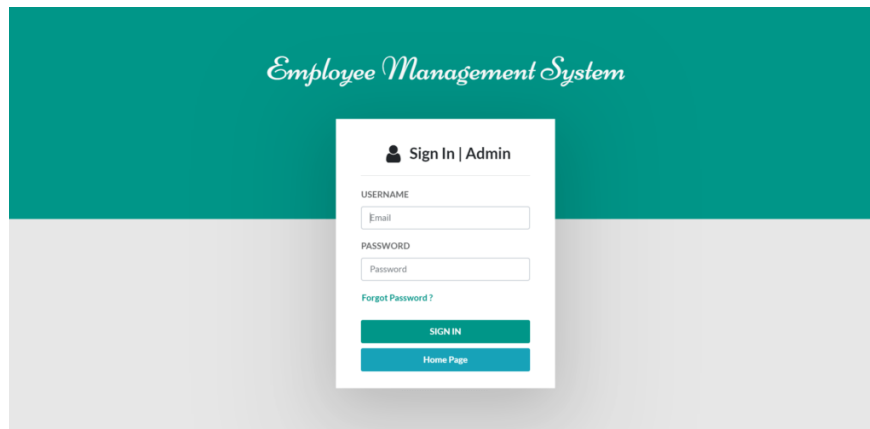
The Employee Management System was implemented using the ASP.NET MVC framework, ensuring a structured separation between data models, user interface, and application logic. The solution consists of clearly organized folders for **Controllers**, **Models**, **Views**, and **Scripts**, making the system scalable and easy to maintain.

10.1 Admin Panel Features

The backend functionality of the admin panel is powered by a series of controllers that handle different modules of the system:

- `AttendanceController.cs`: Manages attendance records, including check-ins, check-outs, and attendance history.
- `EmployeesController.cs`: Handles CRUD (Create, Read, Update, Delete) operations for employee profiles and personal information.
- `LeavesController.cs`: Manages leave applications, approvals, rejections, and leave history.
- `PayrollsController.cs`: Handles payroll generation, salary calculations, and payslip management.
- `ProductivitiesController.cs`: Manages productivity metrics and employee performance tracking.
- `ShiftsController.cs`: Deals with shift scheduling, assigning shifts, and updating shift records.
- `HomeController.cs`: Manages the landing pages and routes for navigation.

These controllers interact with the **Model classes** such as `Employee.cs`, `Attendance.cs`, `Leave.cs`, `Payroll.cs`, and others through the `EMSDbContext.cs`, which acts as the database context using Entity Framework.

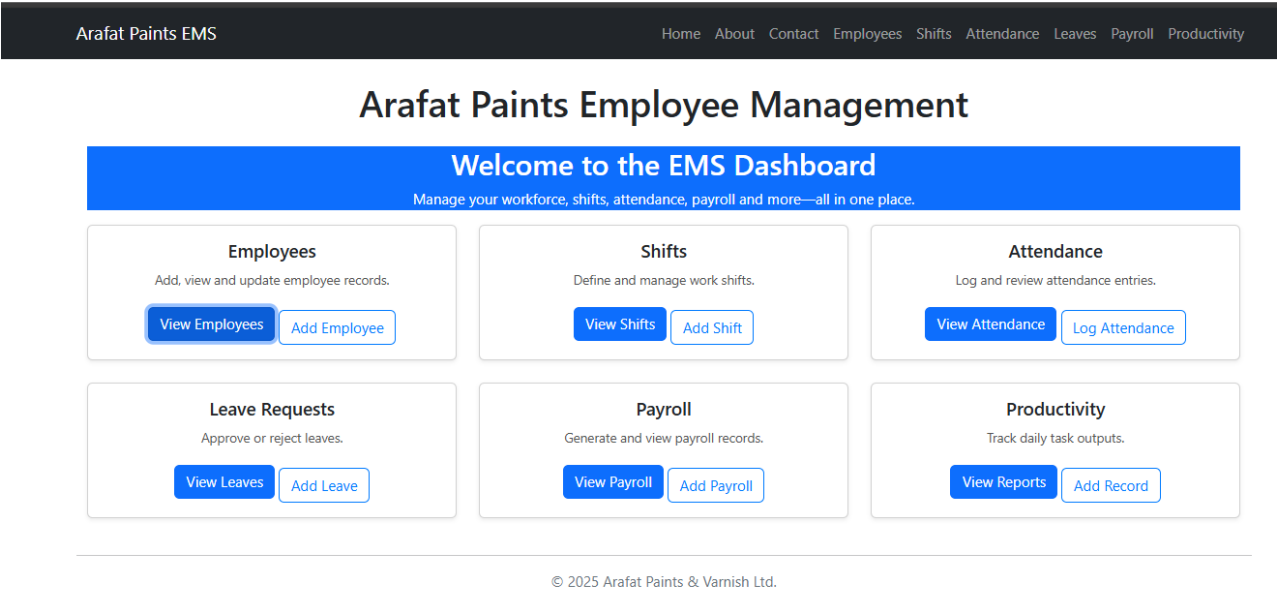


10.2 Employee & User Interface

The **Views** folder contains Razor `.cshtml` files organized by feature (e.g., Attendance, Leave, Payroll, etc.). Each folder inside `Views` corresponds to a module and includes views for listing records, creating entries, editing, and viewing details. These views are dynamic and communicate with controllers to retrieve or submit data securely.

- The `Employees` folder contains views for managing employee profiles.
- The `Attendance` folder handles attendance logs and viewing summaries.
- The `Leave` folder allows users to apply for leave and track status.
- The `Payroll` folder includes salary history, payslip downloads, and payroll details.
- The `Shift` and `Productivities` folders support shift assignments and productivity reports respectively.
- Shared elements like layout and navigation are included in the `Shared` folder.
- The `_ViewStart.cshtml` initializes layout rendering across all pages.

This modular implementation supports efficient development and provides flexibility for future enhancements such as email notifications, report exports, or integration with biometric systems.



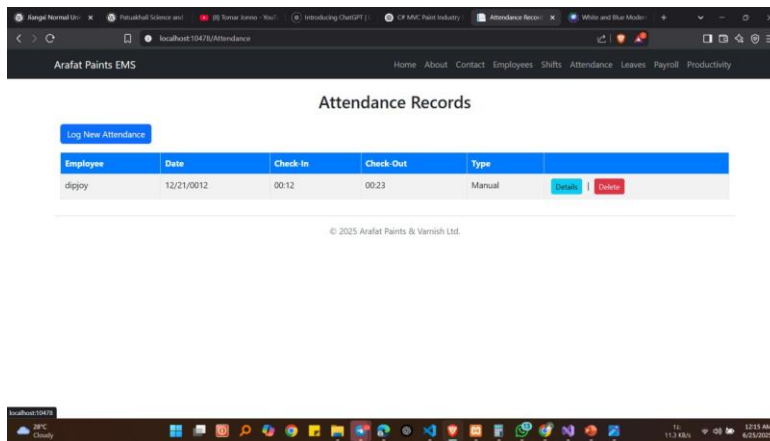
• **Employees:**
The Employees module allows admins to add, edit, view, and delete employee records. It connects to the database through `Employee.cs` and is controlled by `EmployeesController.cs`. All related views are managed inside the `Employees` folder.

Employee List

Create New

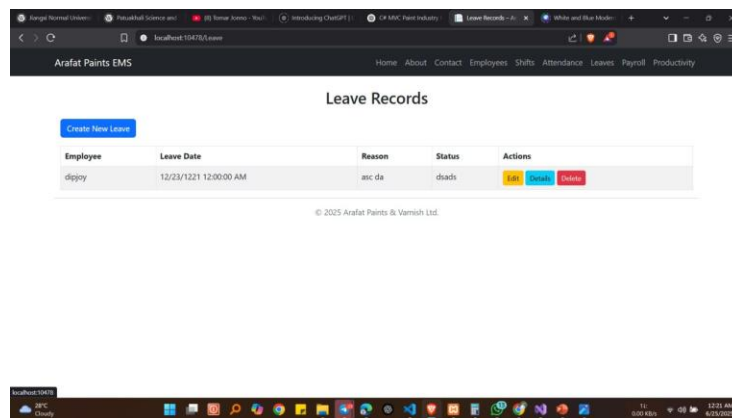
Shift Name	FullName	Designation	DateOfJoining	EndDate	Salary	ContactNumber	Email	Address
EmployeeManagementSystem.Models.Shift?.ShiftName	dipjoy	asa	1211-12-12	N/A	\$22.00	223	abc@gmail.com	23
?ShiftName	sakib	dfir	2001-01-01	N/A	\$12.00	21	asd	asd
EmployeeManagementSystem.Models.Shift?.ShiftName	mahir	adas	2002-01-01	N/A	\$123.00	1231	saC	ADS
EmployeeManagementSystem.Models.Shift?.ShiftName	Shafayet	dsrs	2001-01-01	N/A	\$1,213.00	12321	saC@gmail.com	aerc
EmployeeManagementSystem.Models.Shift?.ShiftName	lala	asa	2001-01-01	N/A	\$22.00	223	saC@gmail.com	ADS
EmployeeManagementSystem.Models.Shift?.ShiftName	nazmul	sakjhadskj	1010-12-01	N/A	\$102,012.00	121201	asash@mail.com	asdkbas
EmployeeManagementSystem.Models.Shift?.ShiftName	asklnsa	as,mndsam	0121-10-20	N/A	\$20,212.00	1065	saaskjnas@mail.com	asdnas
EmployeeManagementSystem.Models.Shift?.ShiftName	asdz	sadsa	1111-12-01	N/A	\$12,012.00	1016516	saC@gmail.com	sada

• **Attendance:**
This module tracks daily employee attendance, including check-in/check-out times. Controlled by `AttendanceController.cs`, it stores records via the `Attendance.cs` model and displays them using views under the `Attendance` folder.



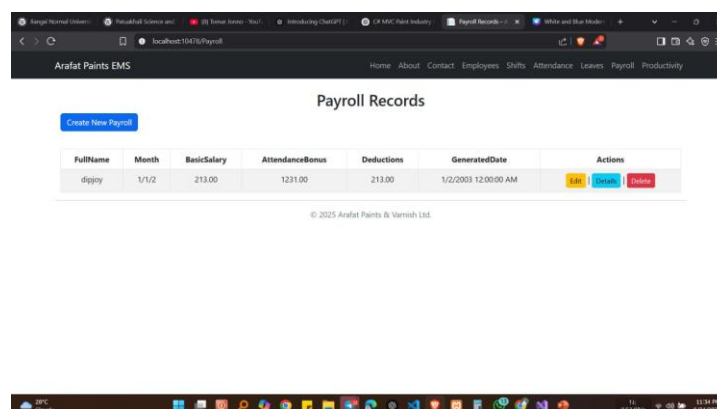
• Leave:

The Leave module handles employee leave requests, approvals, and rejections. It uses `LeavesController.cs` and `Leave.cs` for logic and data storage. Views for applying and managing leave are located in the `Leave` folder.



• Payroll:

This module generates salaries, calculates deductions, and creates payslips. It uses `PayrollsController.cs` and the `Payroll.cs` model. Views for salary processing and history are under the `Payroll` folder.



- **Shift and Productivities:**

Shifts are scheduled and assigned using `ShiftsController.cs` and `Shift.cs`, while productivity tracking is managed by `ProductivitiesController.cs` and `Productivity.cs`. Related pages are in the `Shift` and `Productivities` folders, helping admins monitor work schedules and performance metrics.

11. Testing and Validation

After completing the development phase, the Employee Management System was thoroughly tested to ensure all functionalities worked as intended. Testing was conducted using both manual and functional testing approaches. Each module—such as employee records, attendance, leave management, payroll, and shift scheduling—was tested individually and then integrated together to verify system stability and data flow accuracy. Validation checks were applied to all form fields to prevent incomplete or incorrect data entries. The system was also tested for different user roles (admin, HR, and employee) to ensure proper access control and navigation.

11.1. Test Cases & Results

Test cases were designed for all major features of the system. For example:

- **Employee Module:** Adding a new employee with valid details successfully stores the data; incorrect inputs trigger validation messages.
- **Attendance Module:** When an employee checks in, the current date and time are recorded; duplicate check-ins on the same day are blocked.
- **Leave Module:** A leave request is submitted and appears in the admin panel; when approved or rejected, the status updates in real time.
- **Payroll Module:** Salary calculations correctly reflect attendance and leave deductions; payslips are generated and downloadable.
- **Shift/Productivity Module:** Assigned shifts appear on the employee dashboard; productivity entries are saved and can be reviewed by admin.

All tested features passed under normal conditions. Minor UI alignment issues were identified and fixed. Overall, the system is stable, reliable, and performs well across different user types and scenarios.

Test Case ID	Module	Description	Expected Result	Status
TC1	Employee	Add New Employee	Employee saved successfully	Pass
TC2	Employee	Add Employee with Missing Fields	Validation error displayed	Pass
TC3	Attendance	Mark Attendance (Check-in)	Attendance recorded	Pass
TC4	Attendance	Duplicate Check-in Prevention	Duplicate check-in blocked	Pass
TC5	Leave	Submit Leave Application	Application submitted	Pass
TC6	Leave	Approve Leave Request	Leave status updated	Pass
TC7	Payroll	Generate Monthly Salary	Salary calculated correctly	Pass
TC8	Payroll	Download Payslip	Payslip downloaded	Pass
TC9	Shift	Assign Shift to Employee	Shift assigned	Pass
TC10	Productivity	Add Productivity Record	Productivity stored	Pass

12. Conclusion

The development of the Employee Management System has successfully addressed the key challenges faced by organizations in managing their workforce. By integrating modern technologies such as ASP.NET MVC, SQL Server, and responsive web design, the system provides a robust platform for handling employee records, attendance, leave management, payroll processing, and shift assignments.

This project not only improved the efficiency of routine HR operations but also enhanced data accuracy, security, and accessibility. Through a user-friendly interface and role-based access, both administrators and employees can interact with the system effectively. Real-world validation through testing and industry interviews ensured that the features built into the system are practical and aligned with actual organizational needs.

Overall, the system demonstrates how automation and structured architecture can transform manual processes into efficient, scalable, and intelligent solutions. It lays a strong foundation for future improvements such as biometric integration, analytics dashboards, and mobile app support.

13. References

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